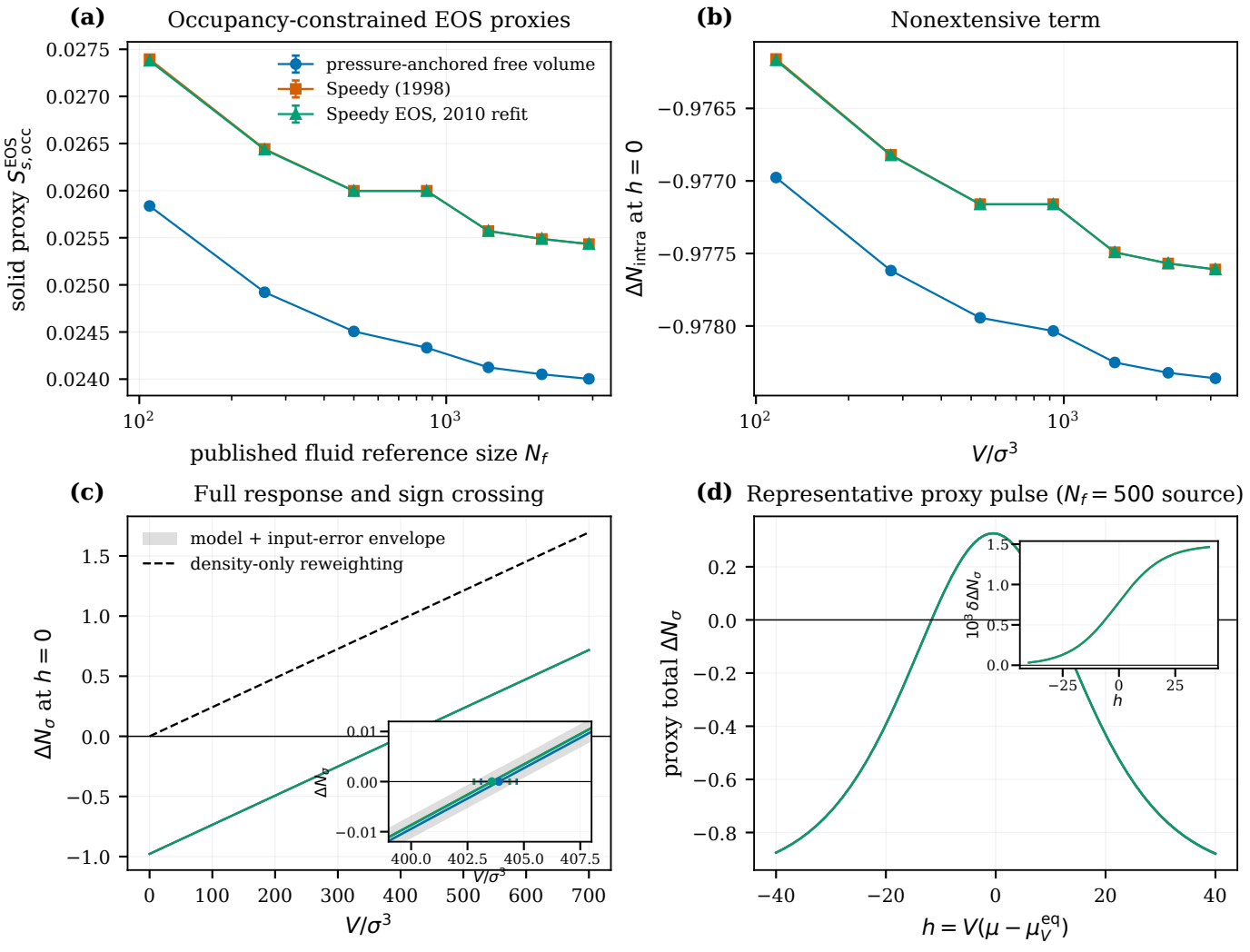

(a)
Occupancy-constrained EOS proxies
pressure-anchored free volume
Speedy (1998)
Speedy EOS, 2010 refit
solid proxy $S_{s,\mathrm{occ}}^{\mathrm{EOS}}$
published fluid reference size N_f
(b)
Nonextensive term
$\Delta N_{\mathrm{intra}}$ at $h = 0$
V/σ^3
(c)
Full response and sign crossing
model + input-error envelope
density-only reweighting
ΔN_σ at $h = 0$
V/σ^3
(d)
Representative proxy pulse ($N_f = 500$ source)
proxy total ΔN_σ
$10^3\,\delta\Delta N_\sigma$
h
$h = V(\mu - \mu_V^{\mathrm{eq}})$